1 mole of copper i.e., 63.5 g
If the entire current produced in the fuel cell ie., 4 F is utilised for electrolysis, then
2× 63.5 i.e., 127.0 g copper will be deposited at cathode.

20. $Ni^{2+}(aq) + 2e^{-} \rightarrow Ni\ (s)$

$Cr^{2+}(aq)+3e^{-} \rightarrow Cr\ (s)$

The above reaction indicates that 2F charge is required to deposit 58.7g of Nickel form nickel nitrate and 3F charge is required to deposit 52g of chromium.

Given that 2.935 gram of Nickel is deposited

$\therefore$ The amount of charge passed through the cell $= \frac{2F}{58.7g} \times 2.935g$

$= 0.1F$

$\therefore$ if 0.1F charge is passed through chromium nitrate the amount of chromium deposited

$= \frac{52g}{3F} \times 0.1F$

$= 1.733g$

21. Given that

$\left[Cu^{2+}\right] = 0.1M$

$E^{\circ}_{Cu^{2+}|Cu} = 0.34$

$E_{cell} = ?$

Cell reaction is

$Cu^{2+}(aq) + 2e^{-} \rightarrow Cu\ (s)$

$$E_{cell} = E^{o} - \frac{0.0591}{n} \log \frac{[Cu]}{\left[Cu^{2+}\right]}$$

$$= 0.34 - \frac{0.0591}{2} \log \frac{1}{0.1}$$

$$= 0.34 - 0.0296$$

$$= 0.31V$$

22. oxidation at anode

$Mg \rightarrow Mg^{2+} + 2e^{-}$ .....(1) $\left(E^{o}_{ox}\right) = 2.37V$

Reduction at cathode

$Ag^{+} + e^{-} \rightarrow Ag$ ..... (2) $\left(E^{o}_{red}\right) = 0.80V$

$\therefore\ E^{o}_{cell} = \left(E^{o}_{ox}\right)_{anode} + \left(E^{o}_{red}\right)_{cathode}$

$= 2.37 + 0.80$

$= 3.17V$

Overall reaction

Equation (1) + 2 × equation $(2) \Rightarrow$

$Mg + 2Ag^{+} \rightarrow Mg^{2+} + 2Ag$

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